

Eine Woche, ein Beispiel

10.5 cohomology of $\mathcal{A}_g$ and $\mathcal{M}_g$

This document aims at a collection of the known results. As usual, I'm not an expert, but often I need to make these documents to clear my brain.

Ref:

[vdG11] Van Der Geer, Gerard. "The Cohomology of the Moduli Space of Abelian Varieties." arXiv:1112.2294. Preprint, arXiv, December 10, 2011. <https://doi.org/10.48550/arXiv.1112.2294>.

Tautological ring R_i

	0	2	4	6	8	10	12	14	...		
R_1	$\mathbb{Q}$ ϕ	$\mathbb{Q}$ 1									
R_2	$\mathbb{Q}$	$\mathbb{Q}$	$\mathbb{Q}$ 2	$\mathbb{Q}$ $2+1$							
R_3	$\mathbb{Q}$	$\mathbb{Q}$	$\mathbb{Q}$	$\mathbb{Q}^2$ 3	$\mathbb{Q}$ $3+1$	$\mathbb{Q}$ $3+2$	$\mathbb{Q}$ $3+2+1$				
R_4	$\mathbb{Q}$	$\mathbb{Q}$	$\mathbb{Q}$	$\mathbb{Q}^2$	$\mathbb{Q}^2$ 4	$\mathbb{Q}^2$ $4+1$	$\mathbb{Q}^2$ $4+2$	$\mathbb{Q}^2$ $4+3$ $4+2+1$	$\mathbb{Q}$ $4+3+1$	$\mathbb{Q}$ $4+3+2$	$\mathbb{Q}$ $4+3+2+1$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$		
R_∞	$\mathbb{Q}$	$\mathbb{Q}$	$\mathbb{Q}$	$\mathbb{Q}^2$	$\mathbb{Q}^2$	$\mathbb{Q}^3$	$\mathbb{Q}^4$	$\mathbb{Q}^5$	...		
A000009 number of distinct partitions	ϕ	1	2	3 $2+1$	4 $3+1$	5 $4+1$ $3+2$	6 $5+1$ $4+2$ $3+2+1$	7 $6+1$ $5+2$ $4+3$ $4+2+1$			

Rmk. 1. $R_i \neq \mathbb{Q}[\lambda_1, \dots, \lambda_i] / (\lambda_1^2, \dots, \lambda_i^2)$

but

$$\text{Gr } R_i \cong \mathbb{Q}[\lambda_1, \dots, \lambda_i] / (\lambda_1^2, \dots, \lambda_i^2)$$

In fact,

$$R_i \cong \mathbb{Q}[\lambda_1, \dots, \lambda_i] / ((1 + \lambda_1 + \dots + \lambda_i)(1 - \lambda_1 + \dots + (-1)^i \lambda_i) - 1)$$

2. In geometry,

$$\lambda_i = c_i(IE) \quad i=1, \dots, g$$

is the Chern class of the Hodge bundle IE .

$$\begin{array}{ccc} IE & & T_0^*A \\ | & & \downarrow \\ \mathcal{A}_g & & [A] \end{array}$$

When we view $R_{g-1} \subset CH_{\mathbb{Q}}(\mathcal{A}_g)$, λ_g vanishes;

when we view $R_g \subset CH_{\mathbb{Q}}(\hat{\mathcal{A}}_g^{\text{tor}})$, λ_g does not vanish.

$\uparrow$
toroidal compactification
in Faltings-Chai.

Chow Rings of $\tilde{A}_g$ [vdG11, 7]

$$CH_{\mathbb{Q}}(\tilde{A}_1) \cong \mathbb{Q}[\lambda_1]/(\lambda_1^2)$$

$$CH_{\mathbb{Q}}(\tilde{A}_2) \cong \mathbb{Q}[\lambda_1, \lambda_2, \sigma_1]/I_2$$

$$I_2 = \left\langle \begin{array}{l} (1+\lambda_1+\lambda_2)(1-\lambda_1+\lambda_2) - 1, \\ \lambda_2 \sigma_1, \\ \sigma_1^2 - 22\lambda_1 \sigma_1 + 120\lambda_1^2 \end{array} \right\rangle$$

$$CH_{\mathbb{Q}}(\tilde{A}_3) \cong \mathbb{Q}[\lambda_1, \lambda_2, \lambda_3, \sigma_1, \sigma_2]/I_3$$

$$I_3 = \left\langle \begin{array}{l} (1+\lambda_1+\lambda_2+\lambda_3)(1-\lambda_1+\lambda_2-\lambda_3) - 1, \\ \lambda_3 \sigma_1, \lambda_3 \sigma_2, \lambda_1^2 \sigma_2, \\ \sigma_1^3 - 2016\lambda_3 + 4\lambda_1^2 \sigma_1 + 24\lambda_1 \sigma_2 - \frac{11}{3} \sigma_2 \sigma_1, \\ \sigma_2^2 - 360\lambda_1^3 \sigma_1 + 45\lambda_1^2 \sigma_1^2 - 15\lambda_1 \sigma_2 \sigma_1, \\ \sigma_1^2 \sigma_2 - 1080\lambda_1^3 \sigma_1 - 165\lambda_1^2 \sigma_1^2 + 47\lambda_1 \sigma_2 \sigma_1 \end{array} \right\rangle$$